

Poisson regression

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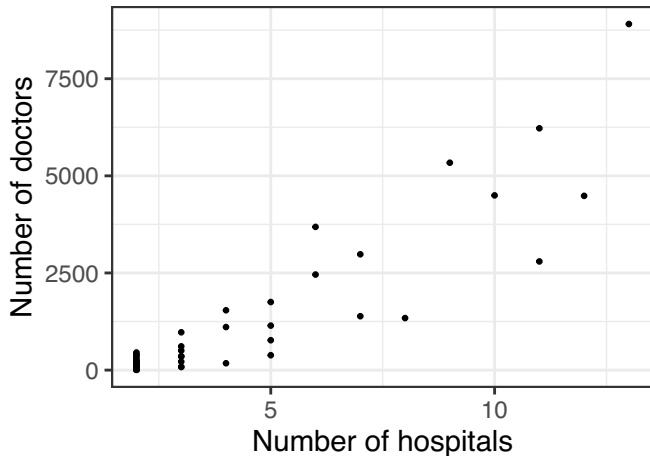
Count variables

Data: Data on medical facilities and doctors from a sample of 53 different counties in the US. Variables include:

- ▶ MDs: the number of medical doctors in the county
- ▶ Hospitals: the number of hospitals in the county

Research question: Can we model the relationship between the number of hospitals and the number of doctors?

Plotting the data

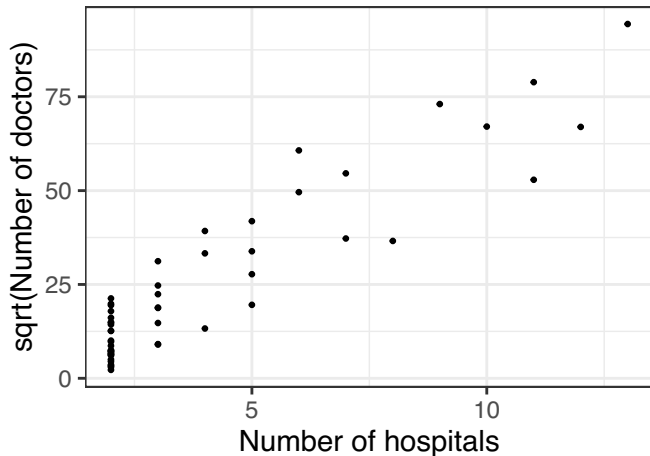


Question: Does a linear regression model seem appropriate for this relationship?

- non-constant variance (sandwich variance? WLS? transformation?)
- discrete response
- might predict negative #s of doctors
- maybe some non-linear relationship? transform?

Trying a transformation

$$\sqrt{Y_i} = \beta_0 + \beta_1 \text{Hospitals}_i + \epsilon_i$$



Is a linear regression model appropriate now?

. shape & constant variance look better

Limitation of transformations

$$\sqrt{Y_i} = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Harder to interpret model in terms of $\sqrt{Y}$
- $\sqrt{\# \text{doctors}}$ is discrete, but $\varepsilon_i \sim N(0, \sigma^2)$ is continuous

$\Rightarrow$ OLS solution is not the same as maximizing log likelihood

$$E[\sqrt{Y_i} | X_i] = \beta_0 + \beta_1 X_i$$

$$\neq \sqrt{E[Y_i | X_i]}$$

Poisson regression

random
component

$$y_i | x_i \sim \text{Poisson}(\lambda_i)$$

$$E[y_i] = \lambda_i$$

$$\text{Var}(y_i) = \lambda_i$$

$$\lambda_i > 0$$

Systematic
component

$$g(\lambda_i) = x_i^T \beta$$

g = link function

Canonical link :

$$g(\lambda_i) = \log(\lambda_i)$$

for Poisson
regression

$$\log(\lambda_i) = x_i^T \beta$$

Fitting the Poisson regression model

```
m1 <- glm(MDs ~ Hospitals, data = CountyHealth,  
          family = poisson)  
summary(m1)$coefficients
```

##	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	5.1168958	0.009801022	522.0778	0
## Hospitals	0.3124424	0.001047882	298.1658	0

Interpreting the Poisson regression model

```
m1 <- glm(MDs ~ Hospitals, data = CountyHealth,  
          family = poisson)  
summary(m1)$coefficients
```

```
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept) 5.1168958 0.009801022 522.0778      0  
## Hospitals   0.3124424 0.001047882 298.1658      0
```

One additional hospital is associated with...
· a increase of 0.31 in estimated
log average # of doctors

· an increase by a factor of $e^{0.31} \approx 1.37$
in estimated average # of doctors
(a 37% increase in estimated average #
of doctors)

Exponential dispersion models

(EDMs)

probability function for Poisson:

$$f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!} = \frac{1}{y!} \exp\{y \log \lambda - \lambda\}$$

Let $a(y, \theta) = \frac{1}{y!}$ (normalizing function)

$\theta = 1$ (dispersion parameter)

$\Theta = \log \lambda$ (canonical parameter)

$k(\theta) = \lambda$ (cumulant function)

$$f(y; \lambda) = a(y, \theta) \exp\left\{ \frac{y \theta - k(\theta)}{\theta} \right\} \quad \leftarrow$$

EDMs: a family of distributions whose probability function looks like this

Activity

Work on the activity (handout).